

E-commerce Sales Performance Report

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1. Project Overview

This project aims to transform raw e-commerce data into actionable business intelligence. By building an end-to-end data pipeline, we have automated the cleaning, storage, and analytical processing of sales information. This report summarizes the key findings derived from the SQL-based analytical layer, focusing on customer segmentation, logistical efficiency, and sales performance.

2. Dataset Summary

The dataset includes transactional records covering:

- Customer Information: Unique identifiers and purchasing behavior.
- Sales Metrics: Revenue, costs, and payment methods.
- Operations: Shipment providers, priority levels, and return status.
- Geographical Data: Country-level distribution of sales.

3. Exploratory Data Analysis (EDA) using Python

Using Python and Pandas, the raw data underwent an ETL (Extract, Transform, Load) process:

- Data Cleaning: Handling missing values and ensuring consistent data types (e.g., converting revenue and costs to numeric formats).
- Transformation: Formatting dates and categorizing transaction statuses to prepare the dataset for relational analysis.
- Validation: Ensuring data integrity before uploading to the PostgreSQL database for advanced querying.

4. Data Analysis using SQL

Our analysis was divided into four core areas to provide a 360-degree view of the business:

- Profitability & Sales Performance: We identified the top-performing categories and monthly revenue trends.

1. Category Profitability Analysis

- Which categories generate the highest total revenue?

- This helps in focusing marketing efforts.

	Category ↑↓🔍	total_revenue ↑↓🔍
1	Furniture	11447320.45
2	Accessories	11284508.23
3	Electronics	11256679.36
4	Stationery	11218219.64
5	Apparel	11194617.38

2. Time-Series Trend Analysis

- What is the total revenue generated for each month of activity?

	month_trunc ↑↓🔍	monthly_revenue ↑↓🔍
1	2025-09-01 00:00:00	118054.47
2	2025-08-01 00:00:00	875412.59
3	2025-07-01 00:00:00	816991.86
4	2025-06-01 00:00:00	787776.40
5	2025-05-01 00:00:00	831412.27

- Customer Behavior & Segmentation: We leveraged Window Functions to rank top products per category and identified "High-Value" customers based on purchase frequency and average ticket size.

3. Customer Behavior (Average Ticket)

- What is the average spending per customer?
- This helps in understanding the value of the user base.

	CustomerID ↑↓🔍	avg_customer_spend ↑↓🔍
1	43562	4889.22
2	93080	4883.34
3	44586	4871.58
4	46660	4853.94
5	44232	4850.51

4. Advanced Filtering: High-Value Customers

- Get the ID of customers who have made at least 5 purchases and have an average purchase spend greater than 500 currency units.

	CustomerID ↑↓▽	total_purchases ↑↓▽	avg_spend ↑↓▽
1	14461	5	2316.91
2	81631	5	1972.12
3	44086	5	1838.35
4	78858	5	1794.19
5	11514	5	1666.97

5. Top 3 most purchased products within each category

- Identifies the best-selling items per category using window functions.

	item_rank ↑↓▽	Category ↑↓▽	Description ↑↓▽	total_orders ↑↓▽
1	1	Accessories	Wall Clock	856
2	2	Accessories	Notebook	833
3	3	Accessories	White Mug	828
4	1	Apparel	USB Cable	848
5	2	Apparel	Wall Clock	825
6	3	Apparel	Desk Lamp	825
7	1	Electronics	USB Cable	859
8	2	Electronics	Wall Clock	843
9	3	Electronics	T-shirt	828
10	1	Furniture	Backpack	878
11	2	Furniture	Desk Lamp	852
12	3	Furniture	White Mug	840
13	1	Stationery	White Mug	843
14	2	Stationery	Office Chair	837
15	3	Stationery	Wall Clock	834

- Logistics & Operational Efficiency: We analyzed shipping costs relative to priority levels and evaluated return rates per product category.

6. Supply Chain Optimization

- Which shipment providers have the highest average shipping costs?

	ShipmentProvider ↑↓	avg_shipping_cost ↑↓
1	FedEx	17.55
2	Royal Mail	17.52
3	DHL	17.45
4	UPS	17.41

7. Order Priority Analysis

- Is there a difference in average shipping costs based on OrderPriority?
- This may reveal if "High" priority orders are more costly to process.

	OrderPriority ↑↓	avg_shipping_cost ↑↓
1	High	17.54
2	Low	17.54
3	Medium	17.37

8. Churn/Returns Analysis

- Which product category has the highest return rate? (i.e., what percentage of sales are marked as "Returned"?).

	Category ↑↓	returned_status ↑↓
1	Furniture	10.03
2	Stationery	9.88
3	Apparel	9.75
4	Electronics	9.69
5	Accessories	9.60

9. Shipping Cost vs. Return Status

- What is the average shipping cost for orders that were "Returned" compared to those that were "Not Returned"?

	ReturnStatus ↑↓	avg_shipping_cost ↑↓
1	Returned	17.46
2	Not Returned	17.49

10. Most used Shipment Provider

- Identify the most frequently used shipment providers and the total volume of completed shipments.

	ShipmentProvider ↕	shipment_completed ↕
1	FedEx	11296
2	Royal Mail	11196
3	UPS	11158
4	DHL	11154

- Geographic & Payment Insights: We mapped sales volume by country and identified the most popular payment methods.

11. Payment Method Performance

- Analyze the average spending and total revenue generated by each payment method to understand customer transaction preferences.

	PaymentMethod ↕	avg_total_revenue ↕	total_revenue ↕
1	Bank Transfer	1267.34	19119073.43
2	paypall	1256.04	18603179.63
3	Credit Card	1253.04	18679092.00

12. Top 5 Countries by Order Volume

- Identify the top 5 countries based on order count and their respective total revenue generation.

	Country ↕	orders_completed ↕	total_revenue ↕
1	France	3820	4768428.71
2	Sweden	3771	4734091.98
3	Germany	3763	4760441.37
4	Netherlands	3753	4588723.21
5	Portugal	3747	4707955.11

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5. Dashboard

- Status: In Development.
- Goal: Visualize the KPIs identified in the SQL analysis (specifically focusing on Revenue by Month, Category Return Rates, and Geographic distribution).
- Planned Technology: PowerBI

6. Business Recommendations

Based on the data insights, we recommend the following strategic actions:

- **Optimize Marketing Spend:** Prioritize advertising for high-value customer segments identified in our SQL analysis.
- **Logistical Review:** Investigate shipment providers with the highest average shipping costs to identify potential cost-saving opportunities.
- **Inventory Management:** Increase stock availability for the "Top 3" best-selling items in each category to maximize revenue.
- **Retention Strategy:** Focus on categories with high return rates to investigate potential quality control issues or inaccurate product descriptions.